MIL-P-51269
14 May 1968
(See section 6)

MILITARY SPECIFICATION

PHOSPHORUS PENTACHLORIDE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reagent grade phosphorus pentachloride.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in conncection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Phosphorus pentachloride shall be pale yellow fuming crystals.

3.2 Chemical characteristics. Phosphorus pentachloride shall conform to the chemical characteristics of table I when tested as specified therein.

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Table I. Chemical characteristics

Characteristic	Maximum percent by weight	Test paragraph
Insoluble matter	0.005	4.2.4.1
Residue after ignition	0.05	4.2.4.2
Sulfur compounds (as SO_4)	0.020	4.2.4.3
Heavy metals (as Pb)	0.002	4.2.4.4
Iron (Fe)	0.002	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the phosphorus pentachloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling for test. Sampling shall be conducted in accordance with table II. Approximately equal portions of phosphorus pentachloride shall be taken from each container in the sample, composited, and placed in a suitable clean, dry glass-stoppered bottle labeled to identify the lot and containers from which it was taken. The total composited sample shall be no less than 1/4 pound (lb).

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Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

4.2.3 Inspection procedure. The composited sample of phosphorus pentachloride shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows (see 6.5):

4.2.4.1 Insoluble matter. Weigh to the nearest 0.01 gram (g) approximately 20 g of the specimen and dissolve in 200 milliliters (ml) of water. Heat to boiling and allow to stand on a steam bath for 1 hour. Filter through a tared filter crucible weighed to the nearest 0.1 milligram (mg), wash with hot water, dry at 105° ± 2° C for 1 hour, cool, and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A-B)}{W}$$

where: A = Weight of crucible and insoluble matter in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.2 Residue after ignition. Weigh to the nearest 0.01 g approximately 2 g of the specimen into a tared porcelain crucible weighed to the nearest 0.1 mg. Gently ignite the specimen at 600° ± 50° C under a hood to constant weight, cool, and weigh to the nearest 0.1 mg. Calculate the percent residue after ignition as follows:

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$$\text{Percent residue after ignition} = \frac{100\ (A-B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.3 Sulfur compounds (as SO_4). Weigh to the nearest 0.01 g approximately 10 g of the specimen and dissolve in 100 ml of hot water. Add 5 drops of bromine water in excess of the amount of bromine water required to develop a bromine color, and boil for a few minutes. Cool, neutralize with sodium hydroxide solution using methyl orange indicator, add 1 ml of concentrated hydrochloric acid and filter. Dilute the filtrate to 100 ml with water, heat to boiling, add 5 ml of 12-percent barium chloride solution, and allow to stand overnight. Filter any precipitate, wash thoroughly, ignite to constant weight in a tared platinum crucible weighed to the nearest 0.1 mg, cool, and weigh to the nearest 0.1 mg. Calculate the percent sulfur compounds as follows:

$$\text{Percent sulfur compounds} = \frac{41\ (A-B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.4 Heavy metals (as lead).

(a) Solution A. Weigh 5.000 g of the specimen and dissolve in 40 ml of water. Boil the solution for 2 minutes, cool, and dilute to 50 ml with water. (This solution is also used in 4.2.4.5.)

(b) Procedure. To 15 ml of solution A, add 4.0 g of sodium acetate and dilute with water to 40 ml. Prepare a control by taking 5 ml of solution A and adding 0.02 mg of lead ion, 7 ml of 10-percent hydrochloric acid, and 4 g of sodium acetate and diluting to 40 ml with water. Add 10 ml of hydrogen sulfide water to both the specimen solution and control. Any brown color produced in the specimen solution shall be no greater than that produced in the control.

4.2.4.5 Iron. To 10 ml of solution A (see 4.2.4.4), add 1 ml of hydrochloric acid and approximately 50 mg of ammonium persulfate crystals and dilute to 15 ml with water. Quantitatively transfer the solution to a separatory funnel, add 15 ml of butanolic potassium thiocyanate solution, and shake for 30 seconds. Any red color produced in the clear butanol (upper) layer shall be no darker than that produced when 0.020 mg of iron is tested in the same manner as the 10 ml of solution A.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Five lb ± 0.5 ounce of phosphorus pentachloride shall be packaged in a type I, class 1 glass bottle utilizing inclosure b (bottles in metal cans) in accordance with the requirements for level A packaging as specified in PPP-C-301, using outer seal A band or tape of PPP-C-186.

5.1.2 Level C. Phosphorus pentachloride shall be packaged to provide adequate protection against contamination, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Eight 5-lb bottles of phosphorus pentachloride, packaged as specified in 5.1.1, shall be packed for shipment in nailed wood box conforming to class 2, style 4-1/2, for a type 2 (average) load of PPP-B-621. Bottles shall be packed in a two bottle by four pattern. The box shall be closed and strapped using galvanized strapping as specified in the appendix of PPP-B-621.

5.2.2 Level B. Phosphorus pentachloride shall be packed as specified in 5.2.1 except that the box shall conform to either a class 1, style 4-1/2 wood box, for a type 2 (average) load of PPP-B-621, or a style optional wirebound box, for class 2 use, for a type 2 (average) load of PPP-B-585. The filled box shall be closed and reinforced using galvanized strapping in accordance with the requirements in the appendix of the applicable box specification.

5.2.3 Level C. Phosphorus pentachloride, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Department of Transportation regulations or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to the special marking specified in 5.3.1 and any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Department of Transportation regulations and MIL-STD-129 (see 6.2).

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following information:

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(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Analysis or maximum limits of impurities.
(e) Precautionary marking:

HAZARDOUS DUST
CAUSES BURNS

Do not breathe dust.
Do not get on skin or in eyes.
Wear rubber gloves and goggles.
In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.
Open container carefully.
Do not add water to contents while in a container because of violent reaction.
Use in a well ventilated area.

5.4 Palletization. When specified in the contract or order, shipping containers of 5.2 shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the hardwood, 4-way, 40- by 48-inch, double-wing pallet of MIL-P-15011 (see 6.2).

6. NOTES

6.1 Intended use. Phosphorus pentachloride is intended for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Special marking other than 5.3.1, if required.
(d) Palletization, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Test precaution. Care must be used in adding phosphorus pentachloride to water because of the great amount of heat produced.

6.6 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-191, dated 25 October 1952.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0441

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
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ATTN: SMUEA-TSE-DD
EDGEWOOD ARSENAL, MD. 21010

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PLATE NO. 15419 (BACK)

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